



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTTP/OPERATION/2025-26/LMI_TPF/299106

Date:03/08/2025(09:47:11)

Subject:LMI for Handling Total Power Failure in Coal Station

LMI for the DOC No. OGN-OPOS-STST-066 , titled "Guidelines for Handling Total Power Failure in Coal Station" Dt.21.05.2025 is prepared and sent for approval

Submitted by:

राजेश कुमार पाढ़ी - सहायक प्रबंधक

RAJESH KUMAR PADHY - 035245 (ASSISTANT MANAGER ,OPERATION) ,Simhadri
(RAJESHKUMARPADHY@NTPC.CO.IN, 7978666574)

On:03/08/2025(09:47:11) **Ref:**SMTTP/OPERATION/2025-26/LMI_TPF/299106

Note No:2

Reviewed.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)
(SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri

On:05/08/2025(20:56:54) **Ref:**SMTTP/OPERATION/2025-26/LMI_TPF/299106

Note No:3

Please review

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:06/08/2025(16:24:18) **Ref:**SMTTP/OPERATION/2025-26/LMI_TPF/299106

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It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:07/08/2025(11:28:55) **Ref:**SMTTP/OPERATION/2025-26/LMI_TPF/299106

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Submitted by:

राजेश कुमार पाढ़ी - सहायक प्रबंधक

RAJESH KUMAR PADHY - 035245 (ASSISTANT MANAGER ,OPERATION)
(RAJESHKUMARPADHY@NTPC.CO.IN, 7978666574) ,Simhadri

On:17/09/2025(03:41:43) **Ref:**SMTTP/OPERATION/2025-26/LMI_TPF/299106

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Reviewed and found in line with OD.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri

On:20/09/2025(16:01:12) **Ref:**SMTTP/OPERATION/2025-26/LMI_TPF/299106

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Reviewed and found in line with OD.

Submitted by:

सतीश चन्द्र मिश्रा - उप महाप्रबंधक

Satish Chandra Mishra - 078593 (DY. GENERAL MANAGER ,MECH MAINT- BOILER)
(SCMISHRA01@NTPC.CO.IN, 9650872225) ,Simhadri

On:20/09/2025(16:23:17) **Ref:**SMTTP/OPERATION/2025-26/LMI_TPF/299106

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Reviewed and found in line with OD.

Submitted by:

अनिल कुमार मौर्या - अपर महाप्रबंधक

Anil Kumar Maurya - 008631 (ADDL. GENERAL MANAGER ,C & I MAINT)
(ANILKUMARMAURYA@NTPC.CO.IN, 8004940574) ,Simhadri

On:22/09/2025(09:04:10) **Ref:**SMTPP/OPERATION/2025-26/LMI_TPF/299106

Note No:9

May Kindly review

Submitted by:

दिव्य चावला - अपर महाप्रबंधक

Divya Chawla - 007348 (ADDL. GENERAL MANAGER ,MTP)
(DIVYACHAWLA@NTPC.CO.IN, 9425534526) ,Simhadri

On:22/09/2025(11:04:51) **Ref:**SMTPP/OPERATION/2025-26/LMI_TPF/299106

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Reviewed. Found in order.

Submitted for approval.

Submitted by:

भास्कर वी के कुरीचेटी - अपर महाप्रबंधक

Bhaskar V K Kurichety - 008045 (ADDL. GENERAL MANAGER ,MTP)
(KBHASKAR@NTPC.CO.IN, 9445864820) ,Simhadri

On:22/09/2025(17:02:46) **Ref:**SMTPP/OPERATION/2025-26/LMI_TPF/299106

Note No:11

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,MTP)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri

On:23/09/2025(10:40:37) **Ref:**SMTPP/OPERATION/2025-26/LMI_TPF/299106

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LMI is in line with the OGN.

LMI may please be approved.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:23/09/2025(10:48:55) **Ref:**SMTTP/OPERATION/2025-26/LMI_TPF/299106

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Submitted for kind approval, please.

Submitted by:

हेमेन्द्र कुमार वर्मा - महाप्रबंधक

Hemendra Kumar Verma - 006459 (GENERAL MANAGER ,MAINTENANCE)
(HKVERMA02@NTPC.CO.IN, 7091092468) ,Simhadri

On:24/09/2025(09:44:25) **Ref:**SMTTP/OPERATION/2025-26/LMI_TPF/299106

Note No:14

Submitted by:

अहमद कौनैन रज़ा - महाप्रबंधक

Ahmad Kaunain Raza - 006151 (GENERAL MANAGER ,O & M)
(AKRAZA@NTPC.CO.IN, 9407064096) ,Simhadri

On:24/09/2025(11:23:51) **Ref:**SMTTP/OPERATION/2025-26/LMI_TPF/299106

Note No:15

Approved & Submitted by:

समीर शर्मा - कार्यकारी निदेशक

Sameer Sharma - 005041 (EXECUTIVE DIRECTOR ,HEAD OF PROJECT)
(SAMEERSHARMA@NTPC.CO.IN, 9650990022) ,Simhadri

On:24/09/2025(11:50:39) **Ref:**SMTTP/OPERATION/2025-26/LMI_TPF/299106

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

Guidelines for Handling Total Power Failure in Coal Station

LMI/OGN-OPS-SYST-066
Rev:00 MAY-2025

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1.	LMI NAME	Guidelines for Handling Total Power Failure in Coal Station
2.	LMI No.	LMI-OGN-OPS-SYST-066 Rev:00 MAY-2025
3.	Reference Document	COS-ISO-00-OGN-OPS-SYST-066 Rev. No: 0, May. 2025
4.	Superseded Document	Not Applicable
5.	Prepared By	Rajesh kumar Padhy Assistant Manager (Operation)
6.	Checked By	Somnath Bhattacharya AGM (Operation)
7.	Checked By	Malay Kanti Jana AGM (EMD)
8.	Checked By	Navin Kishore AGM (EEMG)
9.	Checked By	Nilanjan Mandal AGM (TMD)
10.	Checked By	Suresh Babu Kolla AGM (BMD)
11.	Checked By	Anil Kumar Maurya AGM (C&I)
12.	Checked By	Jayamol Rachel Varughese DGM (MTP)
13.	Checked By	Hemendra Kumar Verma GM (MAINTENANCE)
14.	Rechecked & Reviewed By	Ahmad Kaunain Raza GM (O&M)
15.	Approved By	Sameer Sharma CGM (Simhadri)

TITLE	Guidelines for Handling Total Power Failure in Coal Station
LMI No.	LMI-OGN-OPS-SYST-066 Rev:00

INDEX

SECTION	TITLE	PAGE
1.	Introduction	4
2.	Scope	4
3.	Reference Document	4
4.	Superseded Documents and Revision Index	4
5.	Definition of Black out or Total AC power supply failure	5
6.	Preparation for Black Out	5-6
7.	Activities to be carried out in case of Black out	7-9
8.	Restoration	10
9.	Associated Document (Annexure-1)	11

TITLE	Guidelines for Handling Total Power Failure in Coal Station
LMI No.	LMI-OGN-OPS-SYST-066 Rev:00

1.0 INTRODUCTION:

In the event of a black out, the initial moments are extremely precious, and it requires the right decision to be taken at first instance for speedy restoration of the system. Any lapse or delay in taking the right initial decision may prove to be very costly. Hence, it is of paramount importance that all the operating personnel of the power station/load dispatch centers and the officials involved in grid operation should be thoroughly conversant with the restoration procedure to minimize the time taken in restoration of the system after partial/total black out and subsequent bringing back of units to grid with minimum or no major break down. This LMI identifies the various actions to be taken to avoid major damage to equipment in case of total power failure in the station.

2.0 SCOPE:

Applicable to all Stations

3.0 REFERENCE DOCUMENT:

Operation directive COS-ISO-00-OGN-OPS-SYST-066 circulated by operation services - NTPC corporate center dated 21/05/2025.

4.0 SUPERSEDED DOCUMENTS:

Not applicable

4.1 REVISION

Not applicable

TITLE	Guidelines for Handling Total Power Failure in Coal Station
LMI No.	LMI-OGN-OPS-SYST-066 Rev:00

5.0 Definition of Black out or Total AC power supply failure:

Black out or total AC Power failure is the condition where total AC Power is not available in a station, stage or in a unit. During this period only DC power through the batteries & emergency AC power through DG sets are available.

The occurrence of total power failure will take place in case of a

- Total AC power failure of a unit – unit black out.
- Partial AC power failure to one or more stages of the station – stage black out.
- Grid collapse and all outgoing feeders and units tripped – station black Out.

6.0 Preparation for Black Out:

6.1. Availability of DG sets:

A) Ensure availability of DG sets with daily checking and periodic load test on opportunity shutdown/overhaul. Ensure healthiness of battery by checking of DG trial without battery charger. Maintain optimum level of diesel oil in DG set. Ensure healthiness of DG sets including coolant levels and engine oil levels in regular intervals and record to be maintained. Define the minimum level at which diesel top-up is required.

B) In case of prolonged running of DG sets, ensure availability of diesel for at least additional 8hrs through oil drums or bowser tankers.

Responsibility - UCE

6.2. Availability of Emergency Equipment:

6.2.1 Ensure availability of emergency equipment like DC EOPs, SOPs, air Motors, diesel fire water pumps etc., with daily checking. Once in a month speed checking of all DC drives to be carried out & record to be maintained.

Responsibility- UCE
and MTP

6.2.2 Periodic load trial of all emergency equipment to be done once in 2 years during overhaul and records to be maintained. Equipment protections, interlocks and alarms to be checked once in a year.

Responsibility – UCE
and C&I

6.2.3 If any emergency equipment requires a permit, the duration of PTW to be restricted to 8 hours only. Prior communication to the head of O&M to be ensured and same to be discussed in morning DPM before issuing PTW. If emergency equipment PTW is not returned in 8 hours it is to be informed to head of O&M. SOP to be prepared for handling of total power failure during non-availability of emergency equipment.

Responsibility – UCE&SCE

TITLE	Guidelines for Handling Total Power Failure in Coal Station
LMI No.	LMI-OGN-OPS-SYST-066 Rev:00

6.3 Black out screen in HMI:

Prepare a page on DCS for total power failure in which status of all emergency drives, DG set & breakers & critical parameters are available.

Responsibility- Head of C&I Maint

6.4 CO2 availability:

Ensure availability of required number of CO2 cylinders for purging of H2 from generator casing in case of seal oil failure. Regular checking of CO2 flash evaporator readiness to be ensured. Ensure supply of CO2 flash evaporator is from emergency bus.

Responsibility- UCE.

TITLE	Guidelines for Handling Total Power Failure in Coal Station
LMI No.	LMI-OGN-OPS-SYST-066 Rev:00

7.0 Activities to be carried out in case of Black out:

- In case of unit Black out follow 7.1.
- In case of stage Black out follow 7.1 & 7.2.
- In case of total station Black out follow 7.1 & 7.3.

7.1 In case of Unit Black out

7.1.1 Priority - 1: Ensure starting of the following drives immediately after unit black out to avoid damage to capital Equipment.

1. DC Seal oil pump
2. Main Turbine DC EOP
3. Main Turbine DC JOP (as per OEM recommended speed)
4. TDBFPs EOPs.
5. DC Scanner Air Fan. (Ensure emergency damper open)
6. All APH air motors running
7. DC lighting came into service in auto.
8. Ensure boiler, turbine and generator in tripped condition (Generator breaker opened & excitation off).
9. Close CW interconnection valve to avoid any emergency in other unit of same stage. If stand-by pump is from same unit. Ensure vacuum breaker opening.
10. Ensure FGD Emergency quench valves in opened condition. If not opened, open manually. Ensure FGD bypass damper got opened.
11. Ensure opening of Emergency cooling water to boiler CC Pumps.
12. Ensure opening of Emergency cooling water drain line of Boiler Feed pumps seal water system. (If provided)
13. Ensure running of fire water pumps.

Ensure unit DG set is started on auto on UES bus under voltage, if not, start manually.

1. If main DG set fail to start, start stand-by DG.
2. Close DG incomer breaker to emergency board (UES) after ensuring normal incomer breaker in open condition. If normal breaker not in open condition, give trip command to normal incomer breaker and close DG incomer breaker.
3. Monitor DG Current and other parameters like voltage and frequency.

Note: - Ensure your designated area (boiler desk/turbine desk) equipment are safe before helping others.

Responsibility- UCE

TITLE	Guidelines for Handling Total Power Failure in Coal Station
LMI No.	LMI-OGN-OPS-SYST-066 Rev:00

7.1.2 Priority-II: Ensure Following AC Auxiliaries after UES Charging from DG Set.

1. Start Main TG AOP (AOP-2) & JOP from Unit emergency bus.
2. Start AC SOP and ensure seal oil DP.
3. Start TDBFPs JOPs & AC LOPs.
4. Start AC Scanner Air Fan.
5. Start APHs Main motors and check that APH air motor stops. Ensure physical running of APHs from local.
6. Ensure all respective AC drives having adequate pressure and normal current and stop respective DC equipment.
7. Ensure all AC Emergency equipment like VEFs, LOPs and AHUs from UES are running.
8. Monitor DG Current and other parameters like voltage, frequency, and diesel level.
9. Ensure AC supply extends to all ACDBs (BVDC, TVDC, ID Emergency Board, AC Emergency Board etc.) from UES and other incomers of the respective boards are in open condition.
10. Close boilers stop valves and all drain valves connected to condenser to avoid steaming in condenser and LPT Diaphragm rupture.
11. Ensure closing of SH & RH Spray block valves.
12. Ensure All MS, CRH & HRH line drains to atmosphere are open (like MS strainer drains etc.).
13. Close main TG & TDBFPs Seal steam supply MOVs.
14. Ensure all feed water heater alternate drip valves to condenser are closed.
15. Ensure power supply to barring gear valve/motor is in ON Condition. Take Main Turbine and TDBFPs on barring.
16. Ensure Supply to all chargers (220V charger, 24 V chargers, UPS supply) are charged from Emergency Bus.
17. Ensure changeover of DC lighting to Emergency AC Lighting.
18. Ensure FGD Bypass Damper opened and all AC Emergency drives (agitators, process water pumps etc.) started. (If Total FGD supply is from one unit, Ensure DG started and FGD Emergency Bus charged from FGD DG.)
19. Give stop command to all HT Breakers, Trip command to all Motor Feeders.
20. Ensure all offsite area Chargers are charged from Emergency source or another unit.
21. Ensure AC system Emergency equipment running (AHU, Screw chiller, CT fans from Emergency Bus).
22. Ensure All Lifts supply extended from UES and all persons are evacuated.

Responsibility- UCE

TITLE	Guidelines for Handling Total Power Failure in Coal Station
LMI No.	LMI-OGN-OPS-SYST-066 Rev:00

7.2 In case of Stage Black out:

In case of stage black out, total AC supply failure will be experienced by Units of the affected stage only. However, many common equipment like Air compressors & offsite equipment may be from affected stage. Under such condition, follow instructions mentioned in section 7.1 along with following instructions.

1. Ensure operation of fire water pumps, if fire water pumps supply is from the same stage ensure operation of Diesel driven fire water pumps.
2. Inform other stages about the supply fail & ask them to take standby equipment available at the other stage of affected equipment like IAC/PAC, DM make up pumps etc.
3. Ensure all Station common systems power sources are charged from other stages and equipment are running.
4. Ensure emergency systems (like cooling/lube water tank make-up) are lined up with alternate source for early restoration of system.

Responsibility- UCE

7.3 In case of Total Station Black out:

In case of complete station black out, follow instructions mentioned in section 7.1 along with following.

1. Ensure operation of diesel driven fire water pumps.
2. Ensure all switchyard line breakers are in open condition and ensure system is ready for restoration once start-up power is available. If any breaker has not tripped, open the breaker after noting down the exceptions.
3. Inform to RLDC/SLDC regarding Station black out immediately and request for arranging start-up power as per approved plan as early as possible. Station specific SOP to be made ready.
4. Monitor SF6 pressure / air receiver tank pressure in Switch yard.

Responsibility- UCE & SCE

TITLE	Guidelines for Handling Total Power Failure in Coal Station
LMI No.	LMI-OGN-OPS-SYST-066 Rev:00

8.0 Restoration:

1. Start-up power requirements:

In the case of total station black out start up power requirement to be communicated to RLDC/SLDC for arranging start-up power as per approved plan. Station specific SOP to be made ready

Responsibility- SCE

2. Sources of start-up power:

Ensure start up power from black start hydro/Gas stations or from adjacent Region as per approved Plan in consultation with RLDC.

Responsibility- SCE

3. Responsibilities for start-up power and grid coordination:

- Co-ordination with grid for restoration of Start-up power

Responsibility- SCE/ ASCE

- Charging of lines, monitoring of bus voltage. Ensure availability of Switchyard maintenance group during start-up power line up.

Responsibility-SCE/Switchyard Engineer

4. Restoration at Switchyard Level:

- Communicate with RLDC/SLDC for start-up power at earliest as per approved scheme.
- Note down all breakers opened out and record relays operated.
- Charge the switchyard bus as per SOP for bus charging after clearance from RLDC and monitor voltage at switchyard Level.

Responsibility-SCE/Switchyard Engineer

5. Restoration at Unit Level:

- Ensure all breakers in switchgear are in open condition and ensure system is ready for restoration once start-up power is available. If any breaker has not tripped, open the breaker after noting down the exceptions.
- Charge respective unit station transformer through switchyard.
- Charge the station bus and unit bus and connected downstream transformers/switchgears.
- Change over supply of EMCC from DG set to normal incomer and stop DG Set.
- Ensure all offsite switchgears are charged from their respective incomers. Ensure all relays are reset in the switchgear.
- Ensure all chargers are charged from AC supply.
- After normalization of supply and start the respective AC Drives and ensure adequate pressure & normal current.
- Switch off all DC/Diesel driven/Emergency drives after starting of AC drives and parameter normalisation.
- After restoration of supply to all switchgears, unit start up procedure may be followed as per integrated start up procedure.

Responsibility- UCE

TITLE	Guidelines for Handling Total Power Failure in Coal Station
LMI No.	LMI-OGN-OPS-SYST-066 Rev:00

9. Associated Document

Annexure – I Checklist

S. No.	Description	Status (Yes/No)
1.	DCSOP came in service pressure & current is normal & no drop in H2 pressure. SOP pressure & Current.	
2.	DC EOP & JOP came in service & parameters are normal. Oil pressure & current.	
3.	DC scanner air fan came in service. Ensure emergency damper open.	
4.	Air motors came in auto & ensure APH are rotating at local.	
5.	DG set started in auto & emergency bus got charged.	
6.	Ensure all HOTV/LOTV/HONVs are closed.	
7.	Turbine SVs are closed.	
8.	Generator breaker & excitation system are in off condition.	
9.	Ensure vacuum breaker valve in open condition to avoid LPT diaphragm rupture.	
10.	Isolate seal steam supply to Main TG & TDBFPs.	
11.	Ensure BCW emergency cooling water valves got opened. Keep watching BCW cavity temp.	
12.	If stage blackout or unit blackout has taken place IAC/PAC pressure will drop in other units, start standby IAC /APC from another units / stages.	
13.	Close CW interconnection valve to avoid emergency in other units, if stand-by pump is from same unit.	
14.	Ensure FGD emergency quench valves in opened condition. FGD bypass damper opened on auto.	
15.	Ensure running of fire water pumps	